Pure Function	Impure Function
The return value of the pure functions solely depends on its arguments passed. Hence, if you call the pure functions with the same set of arguments, you will always get the same return values. They do not have any side effects.	The return value of the impure functions does not solely depend on its arguments passed. Hence, if you call the impure functions with the same set of arguments, you might get the different return values For example, random(), Date().
They do not modify the arguments which are passed to them	They may modify the arguments which are passed to them

Now let's see the example of a pure function to determine the greatest common divisor *(gcd)* of two positive integer numbers.

```
let rec gcd a b :=
    if b <> 0 then gcd b (a mod b) else return a;

output
    gcd 13 27;;
    - : int = 1
    gcd 20536 7826;;
    - : int = 2
```

In the above example program *'gcd'* is the name of the function which recursively called till the variable *'b'* becomes *'0'*. Remember *b* and *(a mod b)* are two arguments passed to *'a'* and *'b'* of the gcd function.

1.4.3 Chameleons of Chromeland problem using function

Recall the In the Chameleons of Chromeland problem what you have studied in class XI. suppose two types of chameleons are equal in number. Construct an algorithm that arranges meetings between these two types so that they change their color to the third type. In the end, all should display the same color.

Let us represent the number of chameleons of each type by variables *a, b* and *c,* and their initial values by *A, B* and *C,* respectively. Let a = b be the input property.

The input – output relation is a = b = 0 and c = A + B + C. Let us name the algorithm monochromatize. The algorithm can be specified as

monochromatize (a, b, c)

-- inputs	:	a = A, b = B, c = C, a = b
-- outputs	:	a = b = 0, c = A+B+C

In each iterative step, two chameleons of the two types *(equal in number)* meet and change their colors to the third one. For example, if A, B, C = 4, 4, 6, then the series of meeting will result in

iteration	a	b	c
0	4	4	4
1	3	3	8
2	2	2	10
3	1	1	12
4	0	0	14

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